NYT Business Sentiment Analysis and its Effect on Daily Market Returns: 2015–2024

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What does this mean?

This project tests whether the daily tone of New York Times business news is linked to same-day movements in the stock market.

- Tracked how positive or negative New York Times business articles sounded each day from 2015–2024.
- Analyzed whether days with more upbeat news tend to coincide with higher stock performance.
- In short: tested whether news tone leaves a measurable mark on daily market movements.

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What did I do?

- Collected New York Times business articles and daily stock data
- Scored each article using text-based sentiment analysis.
- Constructed a daily “news mood” index from the article scores.
- Merged daily sentiment with returns for major S&P 500 companies.
- Ran descriptive statistics, correlations, and regression tests.

Where did the data come from?

NYT Business Articles:

- Pulled using the New York Times Archive API (2015–2024).
- Filtered for Business, Business Day, and Economy sections.
- Extracted each article's headline, abstract, and lead paragraph.
- Removed non-business, irrelevant, or duplicate articles.

Sentiment Construction:

- Scored each article using the VADER sentiment model (range –1 to +1).
- Converted raw text into numerical sentiment scores.
- Aggregated all articles per day to create a daily news sentiment index ("news mood").

Market Data (WRDS):

- Pulled firm-level daily returns using the CRSP vwretd variable.
- Selected 13 large S&P 500 companies with full data from 2015–2024.
    - Apple, Microsoft, Amazon - stocks affected by macro data

The New York Times Developers

WRDS WHARTON RESEARCH DATA SERVICES
The Global Standard for Business Research

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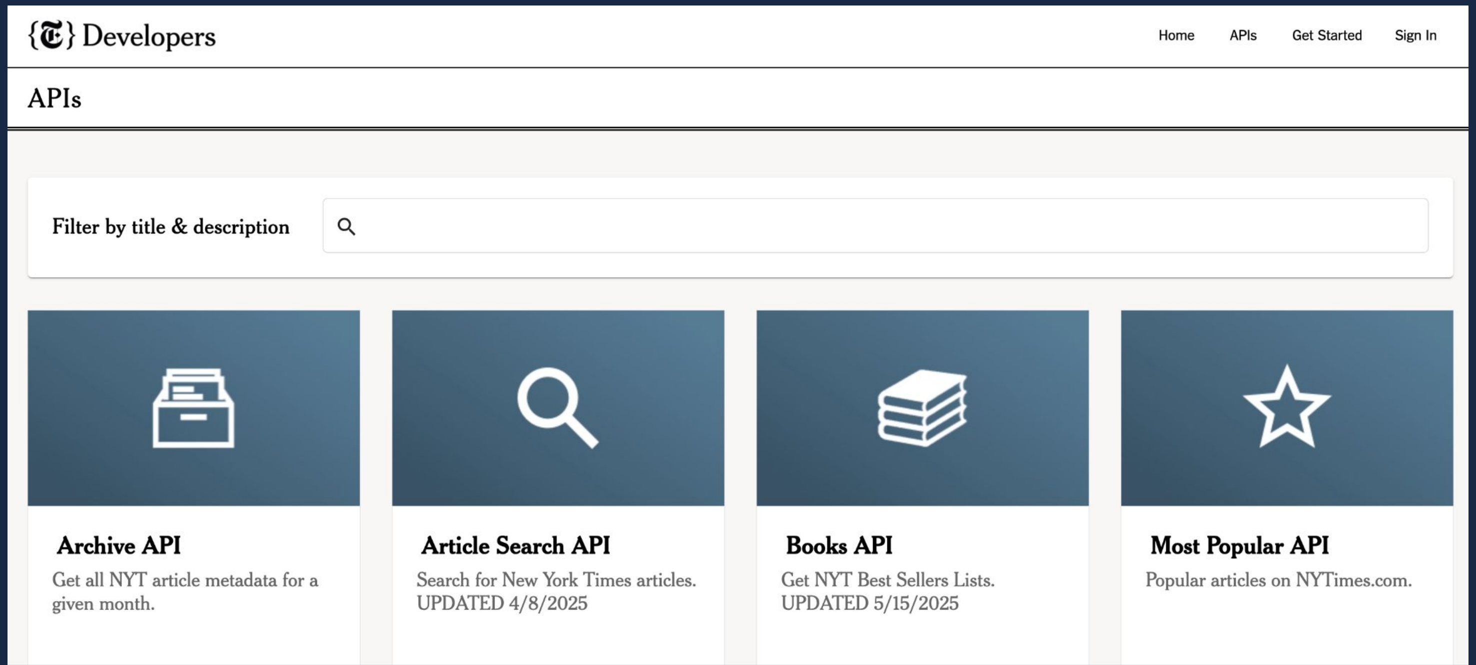
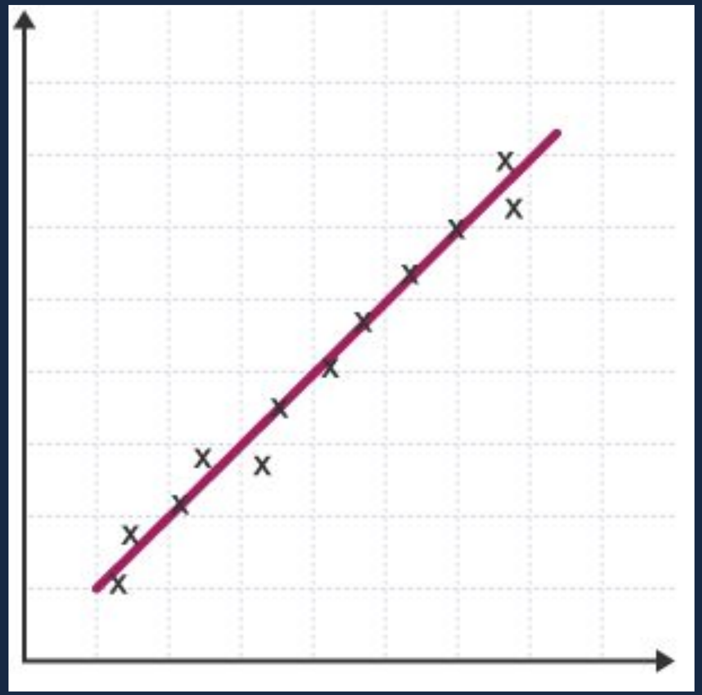


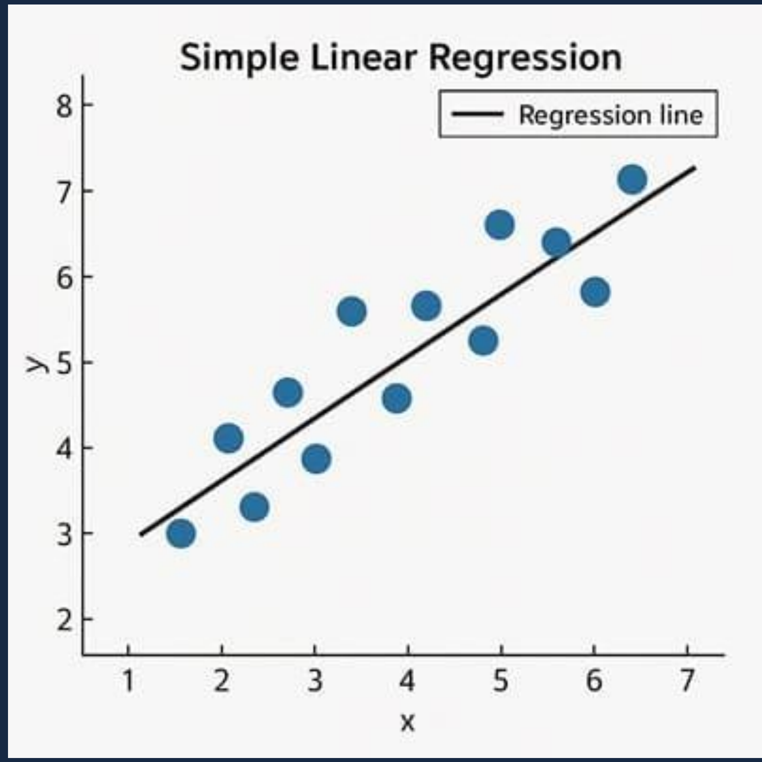
NYT Articles → Sentiment Scores → Daily Index → Merge with Returns → Regression

How the Analysis Was Done

- Merged daily sentiment with company returns into a pooled company-day dataset. (N=33,577)
- Used Python to compute descriptive statistics, correlations, and OLS regressions testing the sentiment–return relationship.

What was I testing?





01 - The Question

- I wanted to see whether the tone of daily business news is connected to how the stock market moves on the same day.

02 - The Test

- I estimated a simple regression model:
  - Daily Return = Intercept + (Sentiment × Coefficient)
- This lets me measure whether more positive news lines up with higher returns.

03 - Why this test?

- Directly measures whether sentiment has any predictive power for daily market movements.
- If the coefficient is large and positive → news tone matters.
- If it's tiny → sentiment barely moves the market.

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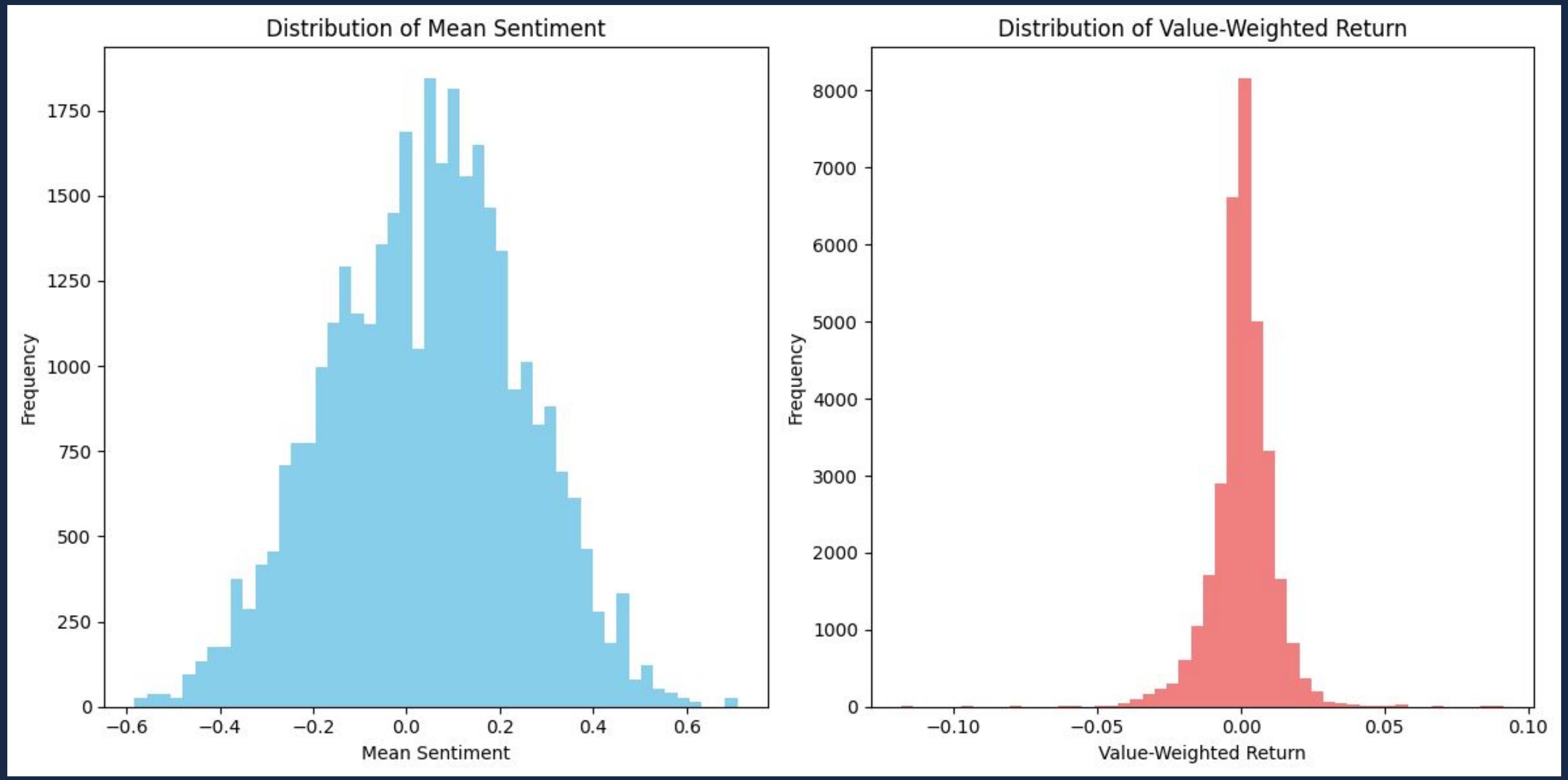
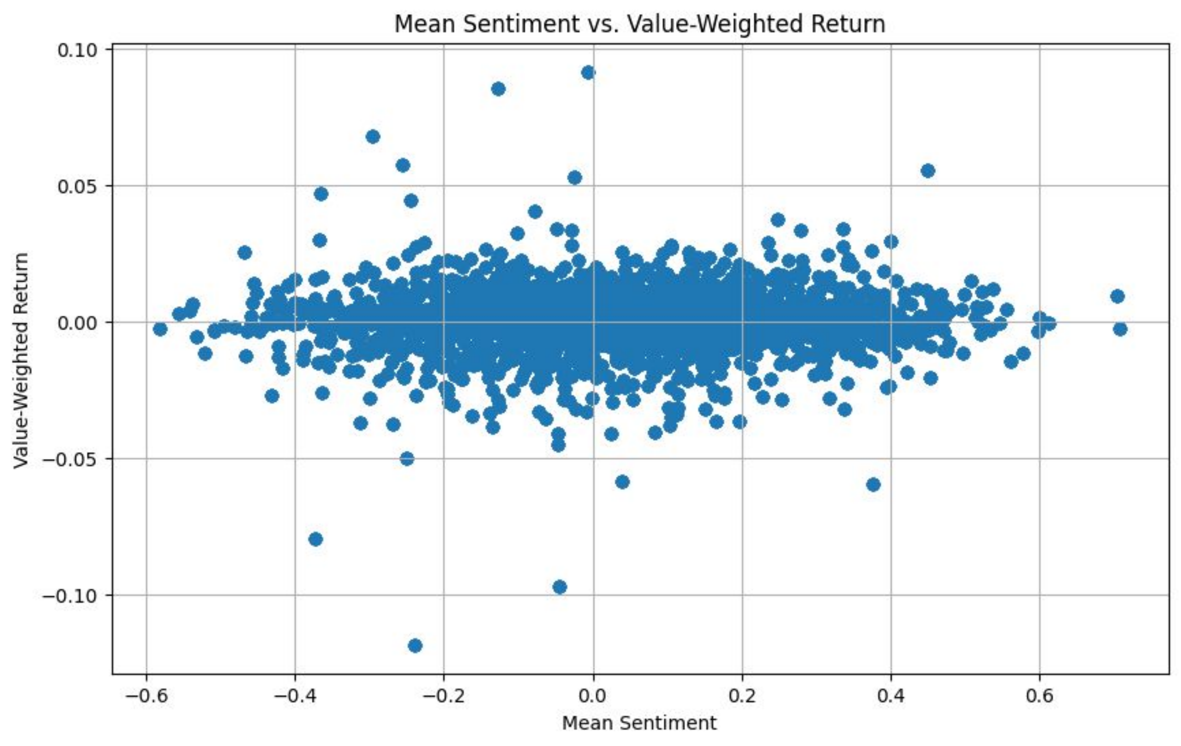
What was my data?





01 - Overall Structure

- Dataset contains 33,577 company-day observations from 2015–2024.
- Each row represents one company on one trading day, paired with my “news mood” score for that date.
- This pooled structure allows many observations per day, giving the analysis more statistical power.

02 - Sentiment Data

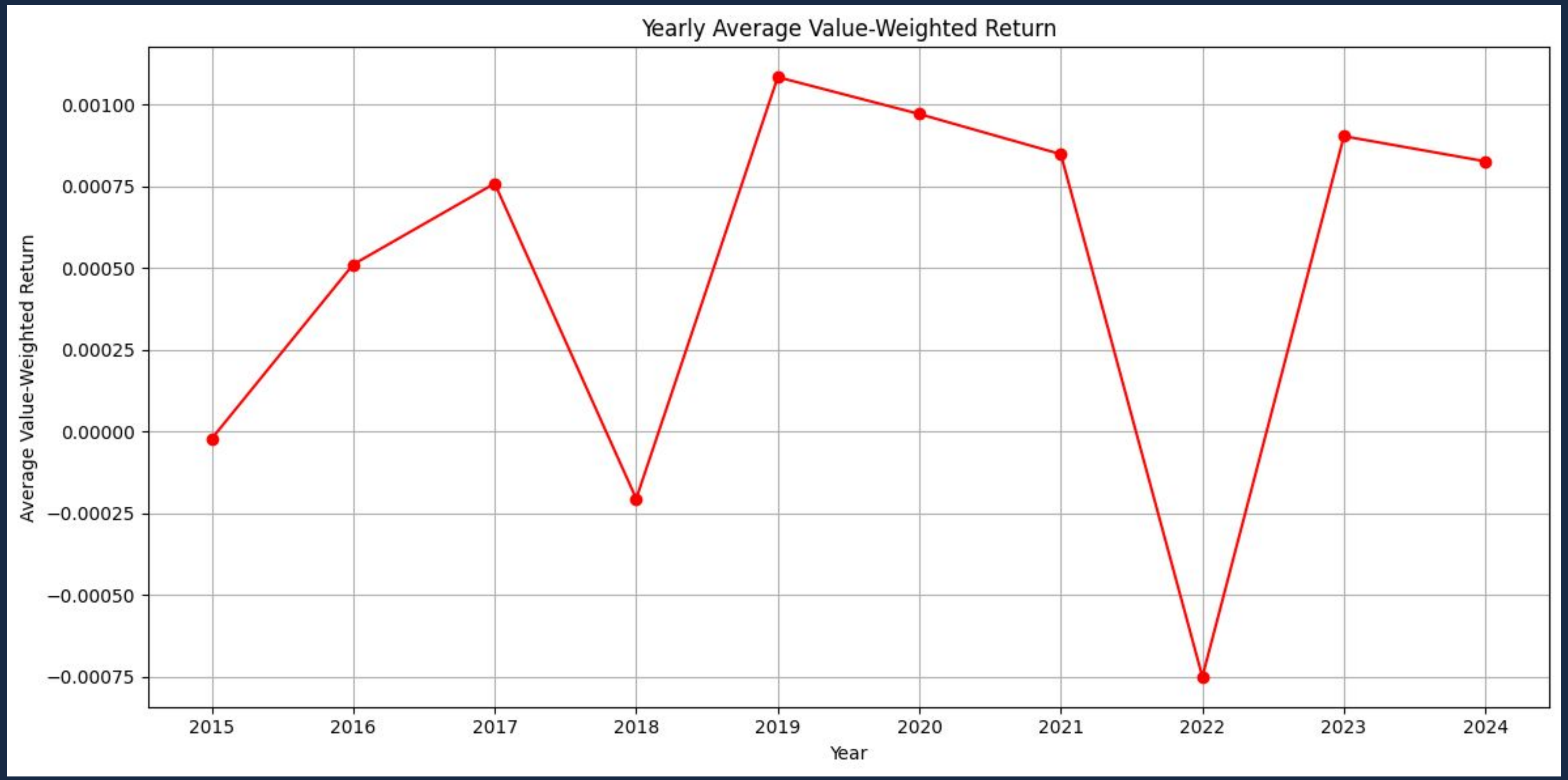
- Daily sentiment scores range from –0.582 (very negative) to +0.709 (very positive), with a mean of 0.042 and a median of 0.0525.
- Most sentiment values cluster near zero, indicating that NYT business articles tend to be neutral in tone on most days.
- The interquartile range (–0.1046 to 0.1840) shows moderate variation across typical days.

03 - Return Data

- Daily stock returns range from –11.82% to +9.16%, with a mean close to zero (0.049%) and a standard deviation of 1.1%.
- The distribution is tightly centered, which is typical for daily equity returns.
- The 5th–95th percentile range (–1.66% to +1.59%) shows that extreme market moves are rare but exist.

Market Behavior over time:

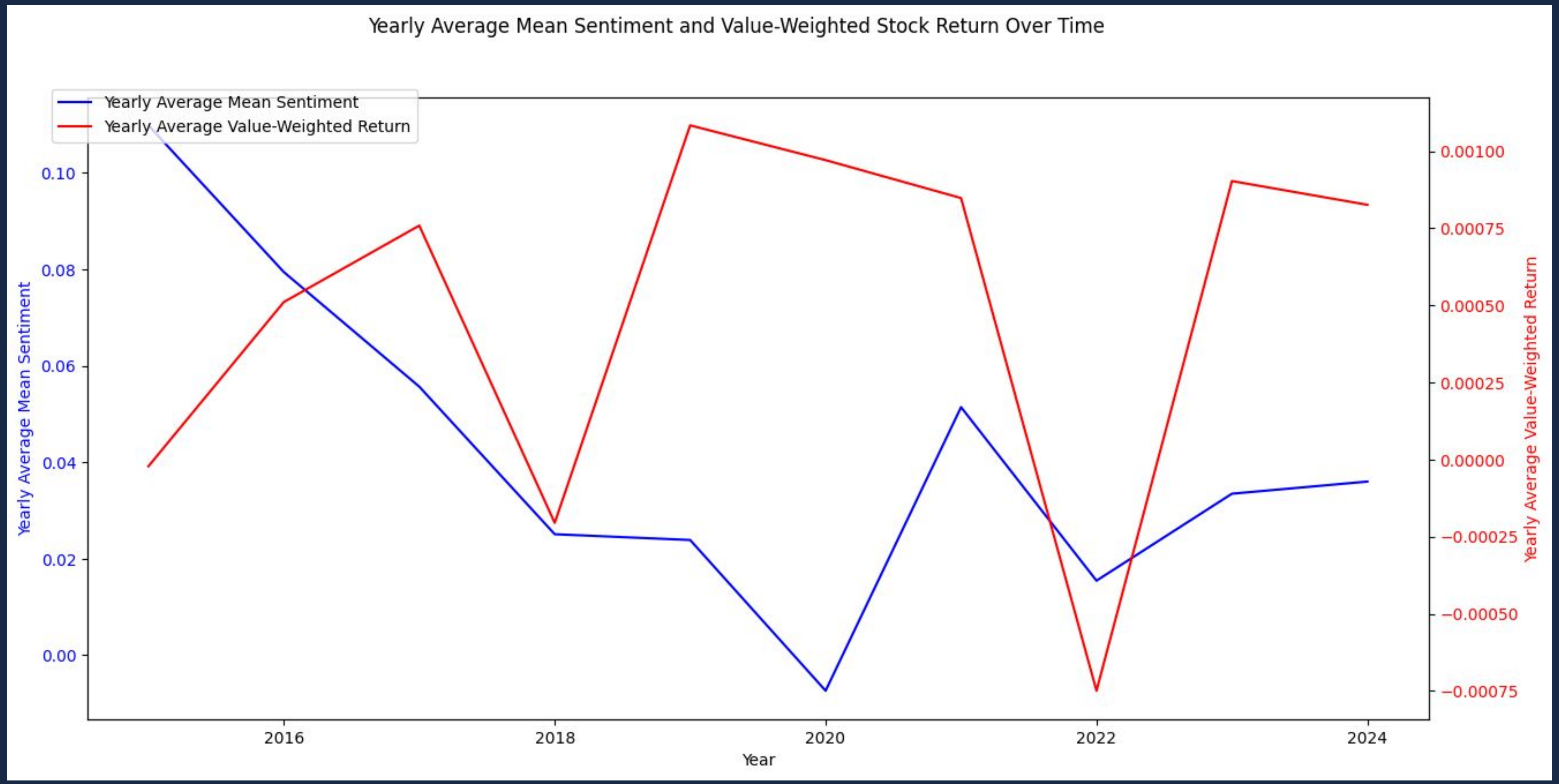
- Returns jump up and down from year to year with no consistent trend.
- Strong years and weak years alternate, showing how unpredictable annual performance is.
- Large swings reflect bigger economic forces, not day-to-day factors like sentiment.



- Daily returns average 0.000491, very close to zero — typical for large-cap U.S. stocks.
- Volatility is modest (Std Dev 0.01107), matching normal S&P 500 behavior.
- Extreme moves happen (–11.8% to +9.2%) but are rare overall.

Sentiment Data Over Time:

- Sentiment is generally slightly positive, with a mean of 0.0420 and median of 0.0525.
- Most values sit near zero (Std Dev 0.2060), meaning tone is usually neutral with moderate variation.
- Extreme sentiment days (–0.58 to +0.71) occur but are relatively uncommon.



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Descriptive Statistics

Key takeaway

- Both sentiment and returns are low-volatility and centered near zero.
- This makes it likely that any relationship will be small.
- These stable distributions make the dataset ideal for regression.

Stat	Mean Sentiment	Daily Returns
N	33,577	33,577
Mean	0.0420	0.000491
Median	0.0525	0.000621
Std Dev	0.2060	0.01107
Min	–0.5821	–0.11817
Max	0.7090	0.09156
5th %ile	–0.3035	–0.01658
25th %ile	–0.1046	–0.00391
75th %ile	0.1840	0.00585
95th %ile	0.3712	0.01586

Regression Results

Intercept: 0.000433

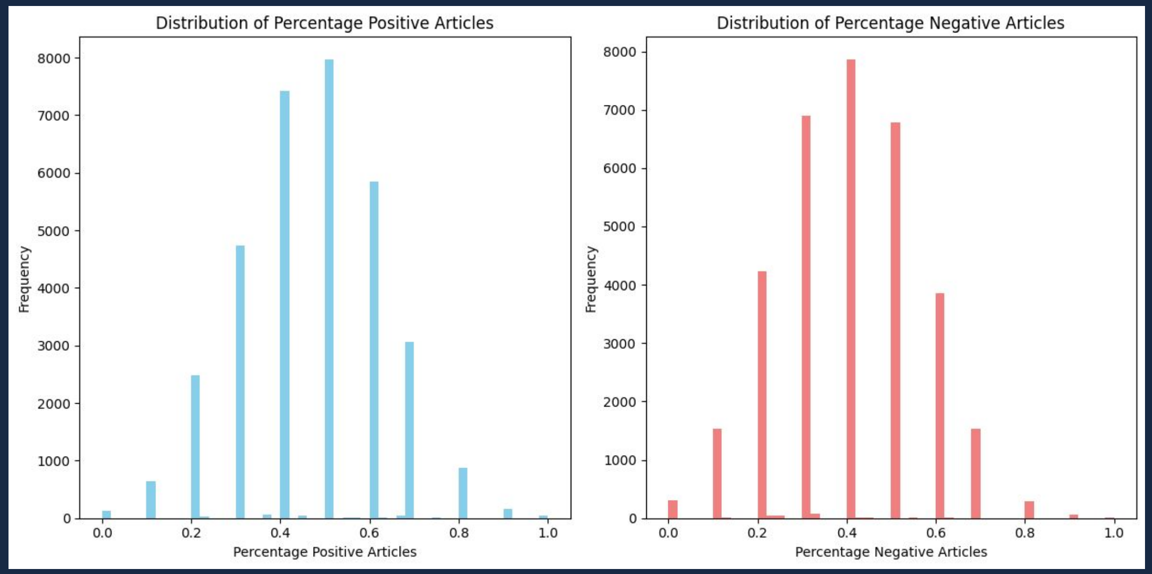
- This is the market's "normal" daily return when the news isn't positive or negative.
- It means the market usually goes up a small amount on an average day, even with neutral news.

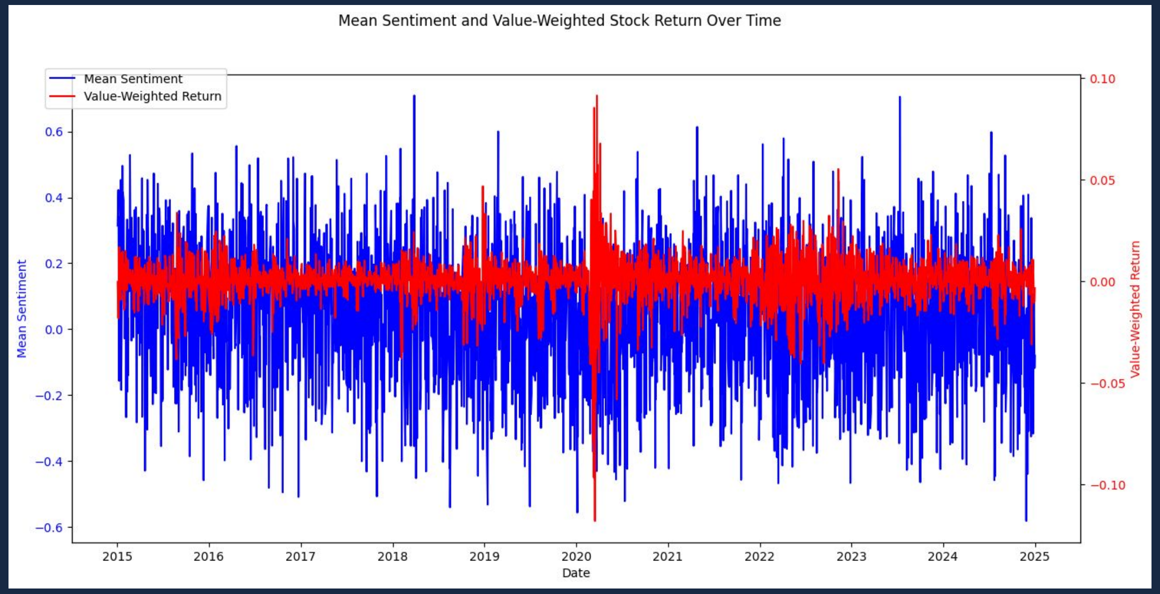
Coefficient on Sentiment: 0.001380

- This shows how returns change when the news tone becomes more positive.
- The effect is extremely small — even big shifts in news tone barely move the market.
- Positive news does line up with slightly higher returns, but the change is too small to matter.

R-Squared: 0.000660

- This shows how much of the market's daily movement can be explained by news tone.
- The answer is: basically none
- Daily stock returns are driven by many bigger factors than how the news sounds.





What were the results?

01 - The Relationship Exists, But It's Tiny

- Positive news days line up with slightly higher returns.
- But the effect is so small that it has no real impact on the market.
- Even a huge shift in sentiment barely moves returns by a few thousandths of a percent.

02 - Sentiment Explains Almost Nothing

- Less than 0.1% of daily return movement is tied to sentiment
- This means news tone has almost zero predictive power

03 - Returns are Driven by Bigger Forces

- Daily market moves depend on earnings, economic data, Fed decisions, and global events.
- These factors overwhelm any small effect from news tone.

04 - Sentiment Reflects the Day

- The sentiment scorer mostly captures the overall mood of the day's events.
- It follows the market narrative rather than influencing market movements
- This means sentiment can describe the day but cannot predict where the market will go next.

Errors & Improvements

01 – Data Source Choice

- I used the New York Times mainly because the API was easy to access.
- More market-relevant outlets (WSJ, Bloomberg) would likely produce a stronger sentiment signal.

02 – Convenience Bias

- The dataset was chosen for convenience, not optimal financial relevance.
- This probably weakened the measured relationship between sentiment and returns.

03 – Coding Process

- My initial code structure wasn't well organized because I relied heavily on AI early on.
- A cleaner, modular workflow from the beginning would have avoided confusion later.

04 – Future Improvements

- Re-run the project with WSJ/Bloomberg headlines, or company-specific news.
- Build a more systematic pipeline: documented steps, functions, and better version control.

Thank You

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